

Prescribing Patterns in the Treatment of Topical Steroid Damaged Face Patients: A Cross-Sectional, Survey-Based Observational Study among Practicing Dermatologists in an Eastern State of India

Abstract

Introduction: Topical steroid damaged face (TSDF) is a common occurrence in the current scenario, where darker - skinned people believe having lighter-colored skin improves their prospects in all aspects of life. This has led to the abuse of topical corticosteroids with unrealistic expectations driven by ignorance of the adverse effects, particularly in the Indian population. As TSDF is virtually an epidemic in India and there is no treatment protocol for dermatologists, we conducted this study to find out what dermatologists prescribe and, in turn, serve as a foundation to develop guidelines in the future. **Materials and Methods:** Forty dermatologists from Odisha, India, were selected to complete the questionnaire, containing 18 questions about diagnosis, treatment and necessary referral for ocular complications. **Results:** All participants were frequently diagnosing and treating TSDF. A total of 92.5% of the participants preferred initiating systemic therapy with isotretinoin or doxycycline as the drug of choice. Clindamycin and tacrolimus were preferred by 55% and 50% of participants, respectively, as the topical drug of choice. A total of 62.5% of the participants advised physical sunscreens and mild cleansers to be used with water at room temperature to wash their face. To reduce facial erythema, brimonidine was advised by 45% of participants. **Conclusion:** Standardized guidelines for managing TSDF are the need of the hour.

Keywords: *Effect of TC on eye, facial redness, TC abuse, TC side effects on face, TSDF*

Introduction

The use of topical steroids as cosmetic and therapeutic creams is sweeping the society as an epidemic. Its prolonged use on the face can cause side effects such as steroid addiction or a topical steroid damaged face,^[1] which can present as an exacerbation of the underlying dermatosis, rebound flaring after discontinuation of topical corticosteroids (TCs), termed 'red skin syndrome', and the appearance of rosacea-like dermatitis induced by chronic TC use. Corticosteroid-induced rosacea-like dermatitis (CSIRD) usually presents as symptomatic facial erythema, edematous inflammation and suffusion.^[2] Managing such cases is nothing less than a herculean task that includes prolonged treatment and proper counseling. The primary treatment for topical steroid damaged face (TSDF) involves discontinuing topical steroids and addressing the exacerbation of symptoms following withdrawal of the drug. Topical emollients, calcineurin inhibitors and

clindamycin are commonly prescribed as off-label indicators. Oral isotretinoin and doxycycline have shown good results.^[3,4] To our knowledge, there are no recommended guidelines for the treatment of patients with TSDF. Hence, there is no uniform prescribing pattern among dermatologists while treating TSDF.

We conducted a cross-sectional questionnaire-based study to analyze the prescribing patterns for TSDF among dermatologists in Odisha, India. The questionnaire was designed based on the existing literature on the management of TSDF.^[3,4]

Materials and Methods

The study was a questionnaire-based, cross-sectional study. The questions were discussed by all the authors as the possible impediments in the treatment of TSDF, and options given were agreed upon by all the authors as the best possible solutions

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in daily practice. The questionnaire was pre-tested on 10 doctors using a pilot study. Modifications were made regarding two questions: “counseling in cases of TSDF” and “any other adverse effects if seen by dermatologists other than those mentioned”. Both the questions were removed because of unanimous decisions by participants in the pilot study. Sixty-four registered dermatologists were randomly selected from a database, and those willing to participate were selected after a written consent at the preliminary stage. The purpose of the study was explained to all participants.

As this was a questionnaire-based study, institutional ethics committee clearance was not required according to current local guidelines.

Forty dermatologists from Odisha who adhered to the existing definition of TSDF^[3] were enrolled and provided the questionnaire via electronic mail, and responses were received similarly. There were 18 questions in total. The first two questions collected information about the awareness of the TSDF entity and the number of cases visiting the doctor in 2 weeks. Seven questions were designed to know the various topical and systemic drugs being used, eight questions were related to supportive therapy, and the last question was to evaluate for ocular complications.

Participants were given 3 weeks to complete the questionnaire and submit their answers along with the duly signed consent form.

Results

The demographic details of the study participants are represented in Table 1.

The responses are mentioned in Table 2.

Discussion

Inappropriate and excessive use of TC on the face is showing a rapidly increasing trend in society and is posing a big menace to the community in the disguise of skin lightening and fairness formulations. Common causes of this trend are the obsession to look fairer and, as a daily cream recommended by friends and family, easy availability as an over-the-counter (OTC) product and in varying combinations at chemist counters, the relatively

low cost of the products and rapidly visible results. Most importantly, the lack of awareness of the potential adverse effects of TC drives its unregulated use.

TSDF is defined as the semi-permanent or permanent damage to the skin of the face precipitated by the irrational, indiscriminate, unsupervised or prolonged use of TC, resulting in many cutaneous signs and symptoms and psychological dependence on the drug. This term was proposed in 2007 and was established in March 2008 by Lahiri and Coondoo.^[3,4]

This entity is associated with the chronic application of TC on the face manifesting as photosensitivity, diffuse erythema and papular and pustular eruptions. It may also be associated with pruritus, burning sensation and visible scaling. The rebound phenomenon in the form of severe flare persists for several weeks. Due to repeated and prolonged use of TC, there is skin atrophy and vasoconstriction. Withdrawal of TC leads to vasodilation and flare related to the release of nitric oxide from the blood vessel endothelium.^[5] The withdrawal flare persists for several weeks if, in particular, the face is affected due to thin skin and high vascularity.

For appropriate management of TSDF, pharmacological and non-pharmacological treatments and counseling are pertinent. The treatment initiation focuses on the complete stoppage of TC.^[6]

In the present study, all participants were well aware of and regularly diagnosed TSDF cases.

75% of the participants advised step-down therapy with low-dose systemic steroids, and most preferred stopping steroids in 2–3 weeks. All participants strictly avoided TC. While many prefer a step-down line of management, it is still controversial.^[7,8]

92.5% of participants preferred to use systemic drugs, and the drug of choice was either isotretinoin or doxycycline. The use of tetracyclines as an anti-inflammatory agent in managing CSIRD has been found to be highly effective and well documented.^[2] Anecdotal experience supports the use of isotretinoin due to its sebosuppressive, comedolytic and anti-inflammatory effects,^[9] which is believed to counter the pathogenesis of TSDF.

55%, 50% and 27.5% of the participants preferred clindamycin, tacrolimus and metronidazole, respectively. Topical clindamycin has been used most widely for acne, and due to its anti-inflammatory properties, it is believed to reduce symptoms of TSDF. Though *Cutibacterium acnes* is considered to be resistant to metronidazole, Khodaeiani *et al.* reported that topical metronidazole 2% gel produced highly satisfactory results in the treatment of acne. The authors reported that the anti-inflammatory, immunosuppressive, inhibition of free-radical generation by human neutrophils and antipruritic properties of

Table 1: Demographic details of the participants (n=40)

Age	32–75 years (Range)	45.72±10.96 (mean±SD)
Male:Female	1.35:1	
Years of Experience	5–48 years (Range)	18.7±10.96 (mean±SD)
Type of Practice	Private practice: 3 Private Hospital: 18 Government Hospital: 19	

Table 2: Results of the survey

Q.no.	Question	Options	No. of responses (n)	Percentage of responses (%)
1.	Do you diagnose TSDF?	All participants were aware and/or diagnosing TSDF.		
2.	What is the average number of patients with TSDF, you see in a week, in routine practice?	1-5 6-10 11-15 16-20 >20	13 10 8 7 2	32.5 25 20 17.5 5
3.	Do you ask your patient to abruptly stop steroid application or do you step down?	Abruptly stop Step down	10 30	25 75
4.	Do you reduce your doses by prescribing a decrease in systemic steroid or lesser potent topical steroid? If yes, then the duration of step-down therapy?	1-2 weeks 2-3 weeks 3-4 weeks >4 weeks	10 11 5 4	33.3 36.7 16.7 13.3
5.	Do you prescribe systemic drugs?	Yes No	37 3	92.5 7.5
6.	What is the systemic drug you choose?	A A, B A, B, D A, C, D A, D B B, C, D B, D C E	10 13 1 1 1 7 1 1 1 1	27 35.1 2.7 2.7 2.7 18.9 2.7 2.7 2.7 2.7
7.	What is the dose of systemic drug that you prescribe?	Variable doses of systemic drugs were used by the participants taking patients' clinical condition and personal experience into account.		
8.	What is the topical drug you choose?			
	a. Tacrolimus	A	20	50
	b. Pimecrolimus	B	5	12.5
	c. Clindamycin	C	22	55
	d. Azelaic acid	D	7	17.5
	e. Metronidazole	E	11	27.5
	f. Benzoyl peroxide	F	2	5
	g. Vitamin D analogues	G	0	0
9.	Do you recommend sunscreens?	Yes No	25 15	62.5 37.5
10.	What is the preferred type of sunscreen?	All the participants advise physical sunscreens.		
11.	Do you prefer emollients?	Yes No	29 11	72.5 27.5
12.	Type of emollient of choice?	Participants prescribing emollients preferred the bland emollients.		
13.	Do you prescribe cleansers?	Yes No	25 15	62.5 37.5
14.	Do you prefer a specific type of cleanser?	Participants prescribing cleansers preferred mild non-foaming cleansers.		
15.	Do you prescribe topical brimonidine?	Yes No	18 22	45 55
16.	What do you recommend for washing face: normal/warm/cold water?	Normal Warm Cold	37 1 2	92.5 2.5 5
17.	What is the cosmetic procedure that you recommend?	Chemical peel Microdermabrasion Laser None	2 0 2 36	5 0 5 90
18.	Do you check for ocular complication in your patients with TSDF?	Yes No	12 28	40 60

metronidazole may also have contributed to the results.^[10] Similarly, the same properties could also have a beneficial effect in TSDF. Topical calcineurin inhibitors are used as a replacement therapy for TC, but there are reports of burning and itching sensation at the application site.^[3]

62.5% of participants advised sunscreens and strictly avoided the chemical sunscreens. As TSDF renders the facial skin prone to further photodamage, sunscreens may help to protect the already damaged skin to some extent. Chemical sunscreens are known to cause allergic reactions and are known to be absorbed by the skin.^[11] Thus, applying chemical sunscreen on the already damaged skin might have an additive negative effect.

Xerosis and pruritus are two of the most common manifestations of TSDF. Emollients are recommended as a first-line therapy for such symptoms to initiate skin repair.^[12] Several moisturizer formulations contain fragrance, parabens, propylene glycol and so on to make them more consumer-friendly but end up causing contact dermatitis. So, bland emollients such as liquid paraffin and white soft paraffin are usually prescribed.

Cleansers with a slightly acidic or neutral pH with minimal skin residue and non-ionic surfactants were preferably advised by our study participants. Synthetic detergents and lipid-free cleansing lotions may be advised in TSDF patients.

It was seen that 45% of the participants use brimonidine, which is a highly selective α_2 -adrenergic receptor agonist whose mechanism of action is vasoconstriction of the superficial skin vasculature and, to a lesser extent anti-inflammatory effects.^[13] This property helps reduce extreme erythema and provides symptomatic relief. The cost, unavailability and limited experience deter the use of this molecule.

92.5% of participants advised washing the face with water of room temperature, though theoretically, cold water may act as a vasoconstrictor and help in reducing erythema. Washing with hot or warm water may be avoided as it may aggravate the erythema.

90% of the participants avoided cosmetic procedures according to the responses. There is a lack of literature regarding the benefits of cosmetic procedures in TSDF. The authors believe ample time must be given for the skin to heal before advising procedures on an already damaged face.

Only 40% of the participants recommend an eye evaluation. The authors highly recommend evaluation for ocular complications to avoid early cataracts and glaucoma, among many other adverse effects.^[14]

Conclusion

Since there are no recommended guidelines and practicing dermatologists use their discretion in managing TSDF

patients, the treatment mostly becomes symptomatic. In such a scenario, treatment-related side effects are not uncommon events. The survey reveals a wide diversity in the prescribing pattern. It includes differences in the prescribed drugs and uneven dosing patterns across all drugs. This study is an initiative to highlight the need to bring out a standardized guideline for the management of TSDF patients.

Limitations

This survey only involved practicing dermatologists in the single state of Odisha, India. A more robust methodology could be adopted to draw further meaningful conclusions of the study. The authors could not ensure prior proper knowledge of TSDF management among the participants included in the study.

Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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Conflicts of interest

There are no conflicts of interest.

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